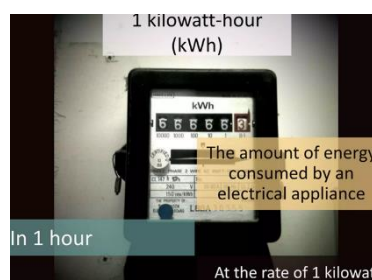


# 22PH202 & ELECTROMAGNETISM AND MODERN PHYSICS

## UNIT I – ELECTRICITY

- Electrical energy
- Capacitor

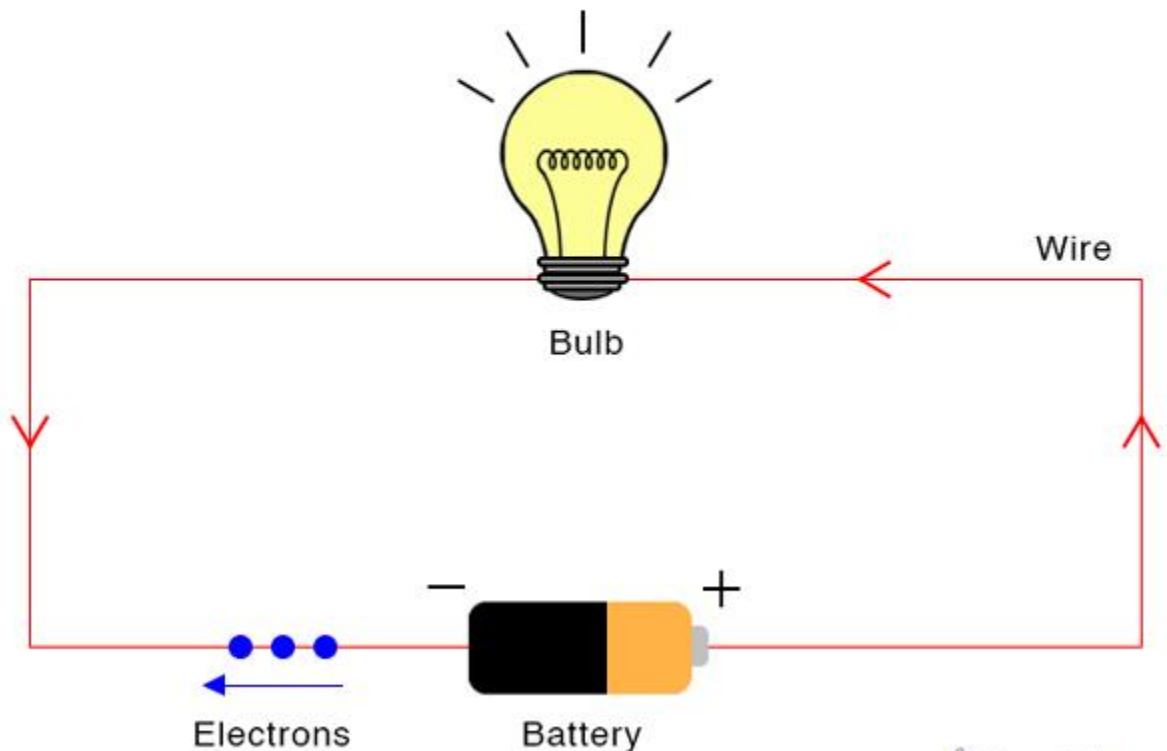


### Introduction

- Electrical energy is a type of kinetic energy caused by moving electric charges.
- The basic unit of electrical energy is the joule or watt-second.
- An electrical energy is said to be one joule when one ampere of current flows through the circuit for a second when the potential difference of one volt is applied across it.
- Examples of electrical energy include batteries, lightning, and electrical charges moving through a wire plugged into the wall socket to power electrical appliances such as televisions and computers.

# Electrical Energy

Electrical energy occurs due to the flow of electrons in a conductor



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## TRANSFORMATION OF ELECTRICAL ENERGY

- **Electricity** is a form of energy that is produced when electrons move from one place to another place.
- **Electrical Energy** can be **transformed** or changed into heat, light, and sound energy, as well as energy of motion.



Heat

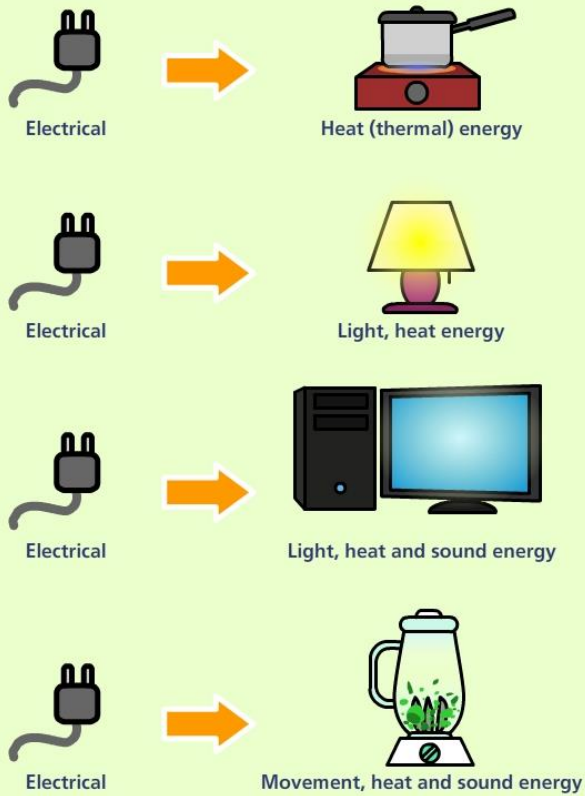


Light

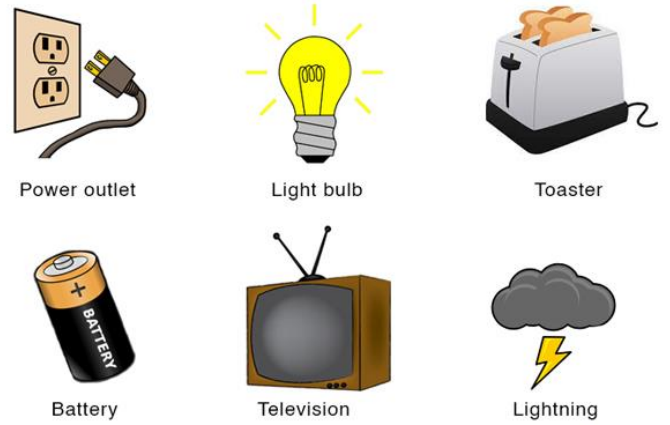


Sound

### Energy Transformations



### Electrical Energy Examples



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## Capacitors

*Capacitors are one of the fundamental passive components. In its most basic form, it is composed of two plates separated by a dielectric.*

*The ability to store charge is the definition of capacitance.*



- Capacitors are **devices** that store electric charge
  - Any conductors can store electric charge, but
  - Capacitors are specially designed devices to store a lot of charge

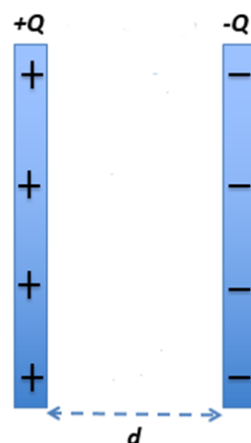


Capacitors are commonly used in a variety of electric circuits

1. To tune the frequency of radio receivers
2. As filters in power supplies
3. To eliminate sparking in automobile ignition systems
4. As energy storing devices in electronic flash units

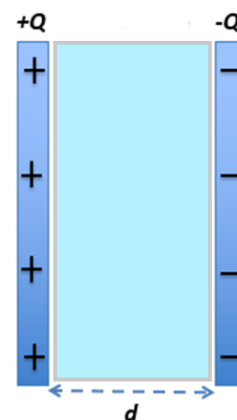
## The capacitance of a parallel plate capacitor

Without a dielectric



$$C = \frac{\epsilon_0 A}{d}$$

With a dielectric



$$C = \frac{\epsilon_0 \epsilon_r A}{d} = \frac{\epsilon A}{d}$$



A capacitor consists of two conductors

When the capacitor is charged, the conductors carry charges of equal magnitude and opposite sign

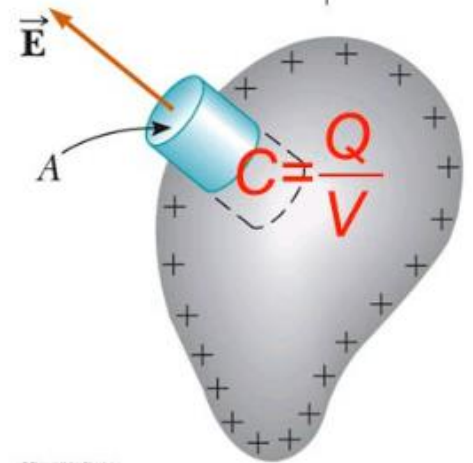
## Capacitance: the definition.

- The **capacitance**,  $C$ , is defined as the ratio of the amount of the charge  $Q$  on the conductor to the potential increase  $\Delta V$  of the conductor because of the charge:

$$C \equiv \frac{Q}{\Delta V}$$

- This ratio is an indicator of the capability that the object can hold charges. It is a constant once the object is given, regardless there is charge on the object or not. This is like the capacitance of a mug which does not depend on there being water in it or not.
- The SI unit of capacitance is the **farad** (F)

$$1\text{F} = \frac{1\text{C}}{1\text{V}}$$



## More About Capacitance

- Capacitance will always be a positive quantity
- The capacitance of a given capacitor is constant
- The capacitance is a measure of the capacitor's ability to store charge
- The farad is an extremely large unit, typically you will see
  - microfarads ( $\mu\text{F}=10^{-6}\text{F}$ ),
  - nanofarads ( $\text{nF}=10^{-9}\text{F}$ ), and
  - picofarads ( $\text{pF}=10^{-12}\text{F}$ )

## Capacitors are in Series:



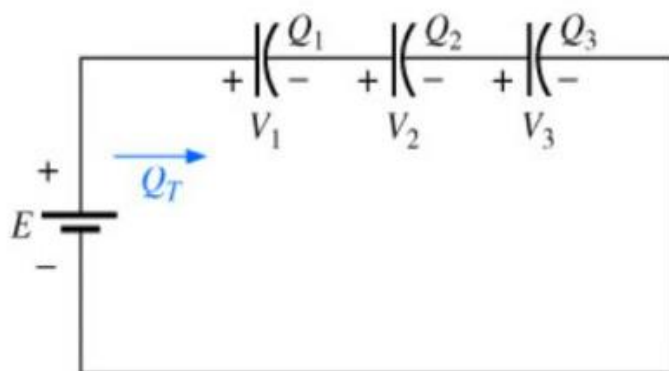
When capacitors are **in series**,  
the charge is the same on each capacitor.

$$V_t = V_1 + V_2 + V_3 \quad Q = CV \Rightarrow V = \frac{Q}{C}$$

$$\frac{Q_t}{C_t} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \frac{Q_3}{C_3}$$

$$Q_t = Q_1 = Q_2 = Q_3$$

$$\frac{1}{C_t} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$



## Capacitors are in Parallel



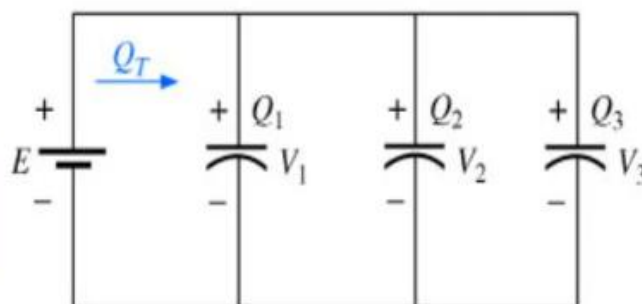
When capacitors are **in parallel**,  
the total charge is the sum of that on each capacitor.

$$Q_t = Q_1 + Q_2 + Q_3 \quad Q = CV$$

$$C_t V_t = C_1 V_1 + C_2 V_2 + C_3 V_3$$

$$V_t = V_1 = V_2 = V_3$$

$$C_t = C_1 + C_2 + C_3$$



### Problem 1:

Find the total capacitance for three capacitors connected in series, given their individual capacitances are 1.000, 5.000, and 8.000  $\mu\text{F}$ .

$$\begin{aligned}\frac{1}{C_s} &= \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \\ &= \frac{1}{1.000\mu\text{F}} + \frac{1}{5.000\mu\text{F}} + \frac{1}{8.000\mu\text{F}} \\ &= \frac{1.325}{\mu\text{F}}.\end{aligned}$$

Now we invert this result and obtain

$$\begin{aligned}C_s &= \frac{\mu\text{F}}{1.325} \\ &= 0.755\mu\text{F}.\end{aligned}$$

Problem 2: Find the net capacitance for three capacitors connected in parallel, given their individual capacitances are 1.0 $\mu\text{F}$ , 5.0 $\mu\text{F}$ , and 8.0 $\mu\text{F}$ .

$$\begin{aligned}C_p &= C_1 + C_2 + C_3 \\ &= 1.0\mu\text{F} + 5.0\mu\text{F} + 8.0\mu\text{F} \\ &= 14.0\mu\text{F}.\end{aligned}$$

### Problem 3:

Calculate the electrical energy consumed by a device with a power rating of 500 watts running for 4 hours.

To calculate the electrical energy consumed, you can use the formula:

Energy (in kilowatt-hours)

=Power (in kilowatts)×Time (in hours)

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Given that the power rating of the device is 500 watts, which is 0.5 kilowatts (since 1 kilowatt = 1000 watts), and the device runs for 4 hours:

$$\text{Energy} = 0.5 \text{ kW} \times 4 \text{ hours} = 2 \text{ kWh}$$

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So, the device consumes 2 kilowatt-hours of electrical energy.

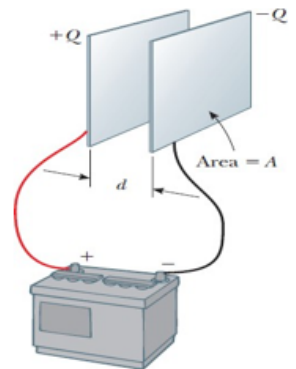
Problem 4:

A capacitor stores charge  $Q$  at a potential difference  $\Delta V$

What happens if the voltage applied to the capacitor is doubled  $2\Delta V$ ?

- (a) The capacitance falls to half its initial value and the charge remains the same
- (b) The capacitance and the charge both fall to half their initial values
- (c) The capacitance and the charge both double

(d) The capacitance remains the same, and the charge doubles



## Summary

- Electrical energy is a type of kinetic energy caused by moving electric charges.
- A capacitor is a device capable of storing energy in a form of an electric charge.